



University of Malawi

CHANCELLOR COLLEGE

PHYSICS DEPARTMENT

1. College	Chancellor College
2. Faculty/School of	Science
3. Department	Physics
4. Programme	BSc (Electronics), BSc (Physics), B.Ed.
5. Module Title	Mechanics and Properties of Matter
6. Course code	PHY 111
7. Level	Year 1
8. Credits	12
9. Lectures per week:	3
10. Lab hours per week	2
11. Tutorial hours per week	
12. Revised	After 5 years
13. Approval Date	

14. Prerequisites (if applicable):
MSCE/GSCE with Physical Science or Physics

15. Co-requisite:
MAT111

16. Module Aims:
To provide the background needed for the further study of University Level Physics. It exposes students to a beginning course in Mechanics and Properties of Matter. Mechanics is broken down into its three components: statics, dynamics and kinematics. These are clearly explained to students so as to bring out their differences and inter-relationship. Students are introduced to the various groups of properties of matter.

17. Intended Learning Outcomes:

On completion of this module the student should be able to:

- a) Explain the concept of velocity, acceleration and vectors.
- b) Describe the force, pressure and gravity.
- c) Describe the Newton's laws of motion.
- d) Solve problems involving Newtons' laws.
- e) Describe work, energy and conservation of energy theorem
- f) Describe temperature and laws of thermodynamics.
- g) Define and describe Heat capacities of materials.
- h) Solve problems involving heat capacity, specific heat capacity and latent capacities.
- i) Solve problems involving gases laws and principle molar heat capacity of gases.

18. Indicative Content

Mechanics

- (a) Position, velocity, acceleration Scalars and Vectors, Dimensional analysis,
- (b) Forces: Gravity, Pressure. Statics, Dynamics Rectilinear, Projectile motion
- (c) Momentum, Newton's 1st and 2nd Laws. Conservation of momentum, collisions,
- (d) Energy and work, Kinetic and potential energy, Conservation of energy

Thermal Properties of Matter

- (a) Thermometry and calorimetry. Temperature, temperature scales
- (b) Liquid in glass thermometers, thermocouple
- (c) Heat capacity, Latent Heat
- (d) Cooling Laws: Newton's Law of cooling, The five-fourths power law
- (e) Gases: Gas Laws, Ideal gas law, First law of thermodynamics
- (f) The principle molar heat capacity of a gas
- (g) $C_p - C_v = R$ for an ideal gas

19. Assessment

40% Continuous Assessment 60% Examination

20. Teaching and Learning Methods / Activities

Lectures, tutorials, practical work, projects work, oral presentations and reading assignments

21. Prescribed Texts

Knight, R. D., Jones, B. and Field S. (2007). College Physics, Prentice Hall.

Nelson M. and Parker P. (1995). Advanced Level Physics, Heinemann International

Serway, R. A., & Faughn J. S., (2000). College Physics (5th ed). Saunders College Publishing.

22. Recommended Texts:

Knight, R.D. (2004). Physics for Scientists and Engineers with Modern Physics: A Strategic Approach, Addison Wesley.

Sears, F. W.; Zemansky, M. W. and Young, H. D. (1991). College Physics (7th ed.) Addison Wesley

Young, H.D. & Freedman R.A. (2004). University Physics. (11th ed), Pearsons, Canada